BRIAN CAO

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Education

Stony Brook University **Expected May 2027**
Majors (B.S.): Computer Science Honors GPA: 3.86; Dean's List
Coursework: Algorithms, Deep Learning, Software Eng., Systems, Linear Algebra, Graph, Probability, Security, Cloud Computing

Experience

Washington University Summer Engineering Fellowship (WUSEF) – CyanoAgent St. Louis, MO
AI Research Fellow **May 2026–Aug 2026**

- Developed CyanoAgent, a database-grounded knowledge graph for an LLM research assistant, achieving 96% accuracy on gene-identification queries vs 4% for a baseline LLM.
- Designed a three-tier trust system (verified, extracted, inferred) to control how facts propagate through the knowledge graph and implemented guardrails that blocked unverified claims from being presented as fact
- Diagnosed and fixed data-resolution bugs causing incorrect gene and organism matches, correcting output accuracy across every supported species.
- Implemented a multi-stage hypothesis generation pipeline that aggregates evidence across 6 databases, leverages Claude Opus to draft structured research proposals, and enforces target-identity validation guardrail.

Stony Brook University Computer Science Department – Operations & Data Stony Brook, NY
Software Engineer and IT Support **Jan 2025–May 2026**

- Automated data processing pipelines in Python for graduate application workflows, reducing manual review time by 60% and improving system maintainability through modular architecture
- Configured and troubleshot 10+ Power Distribution Units (PDUs) for network device monitoring, resolving authentication, firmware, and polling issues to improve infrastructure reliability
- Migrated 5+ departmental websites to Drupal 10 CMS, improving performance and accessibility scores by 40%
- Integrated 20+ campus devices into the Zabbix monitoring system for network monitoring, reducing issue detection time

Pulp Corporation Brooklyn, NY
Software Engineer Intern **Jan 2025–Aug 2025**

- Engineered a full-stack job-matching platform using Next.js, TypeScript, and Neo4j, leveraging graph-based data modeling for backend systems to automate matching for 100+ users and reduce manual review time by 75%
- Implemented data ingestion pipelines for CSV parsing, validation, and transformation with REST APIs, improving data processing efficiency and increasing onboarding speed by 300%
- Built and standardized reusable UI components using design tokens and Storybook, improving frontend system consistency and development efficiency by 200%

Stony Brook University Campus Residences Stony Brook, NY
Resident Assistant **Aug 2024–Present**

- Spearheaded 5 campus-wide events with over 600 attendees collaborating with 10+ RAs, 650% more than average
- Cultivated an inclusive, supportive, and safe residential community for 30+ students through mentorship, conflict mediation, hall engagement initiatives, and collaboration with campus resources.
- Handled and mediated 5 roommate conflicts to ensure the inclusivity inside dorm building

Projects

Stony Brook Academic Advising Bot (Python, Next.js, Supabase, Pinecone, LangGraph) **Mar 2026–Present**

- Built a full-stack advising assistant with a Next.js/Tailwind frontend and Python Flask backend, ingesting degree requirements and policies to streamline student advising workflows
- Built a semantic vector store and retrieval pipeline using Pinecone and Supabase, indexing scraped institutional data to synthesize grounded user resolution.
- Integrated LLM summarization and strict context grounding to deliver concise academic recommendations.
- Automated data ingestion and scraping to reduce manual data lookup time for advisors.

Playlister (React, Node.js, Express, MongoDB) **Aug 2025–Dec 2025**

- Implemented client-side state management to handle complex UI behavior, including undo/redo functionality, edit modals, and responsive components
- Designed RESTful API endpoints using Express and structured MongoDB schemas to support playlist CRUD operations and embedded song data
- Built a full-stack playlist management application with React and Node.js, enabling dynamic user interactions with real-time state synchronization

Technical Skills

Languages: Python, SQL, PostgreSQL, C/C++, Java, JavaScript, TypeScript, R, OCaml, HTML, CSS
Libraries/Tools: AWS, Hugging Face, Transformer Library, PyTorch, Jupyter, Numpy, LangGraph, Bash, Git, Flask, Bootstrap, .NET, LaTeX